



Bubbles

Your AI conversation co-pilot — live coaching while you talk,
and a smart assistant that remembers.

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Final Year Project · Mid Evaluation · BS Computer Science · COMSATS University Islamabad, Lahore Campus

Two people. One split-stack build.

Muhammad Ahmad

FA22-BCS-025 · App · Architecture · Frontend

- Flutter client across 6 platforms
- System architecture & data model
- Auth, UI/UX, caching, releases

Attique Rehman

FA22-BCS-164 · AI · Backend · Memory systems

- FastAPI brain & LLM router
- Knowledge graph + vector memory
- Real-time pipeline & workers

A friend in your ear — and a memory that lasts.

1

Listens

Hears the conversation as it happens, in real time.

2

Suggests

Quietly whispers what to say next, so you never freeze.

3

Remembers

Keeps your people and topics, so you can ask about them later.

Writing got autocorrect. Speaking never did.

Speech happens in real time. We notice the wrong word or wrong tone only after the moment is gone.

We kept losing the moment.

01

The blank-out

Froze mid-interview — knew the answer, couldn't get it out.

02

The forgotten promise

Couldn't recall what a client actually agreed to last week.

03

The walk-home regret

Realized the better phrasing only hours after it mattered.

Two camps today — neither helps in the moment.

Meeting notetakers

Otter · Fathom · Notta · Gong

Transcribe & summarize — but only after the meeting ends.

General assistants

Siri · Google Assistant · ChatGPT

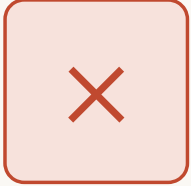
Answer commands — but forget you between sessions.

Three things nobody does together.



No real-time coaching

Everything on the market is post-meeting analysis.



No long-term context

Assistants forget who and what you talked about.

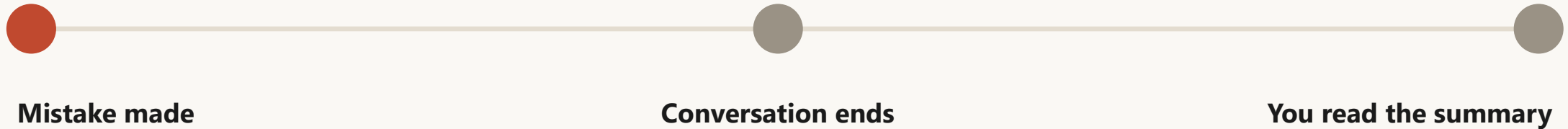


Not privacy-first

Enterprise tools mine your data for the org, not you.

WHY POST-MEETING ISN'T ENOUGH

The value of advice decays in seconds.



Feedback only ever lands here → too late to change the outcome.

PROBLEM STATEMENT

Success often depends on communication, yet errors are hard to catch while speaking. Existing AI only transcribes or summarizes after the fact — with no live, context-aware, privacy-respecting feedback.

WHY IT MATTERS

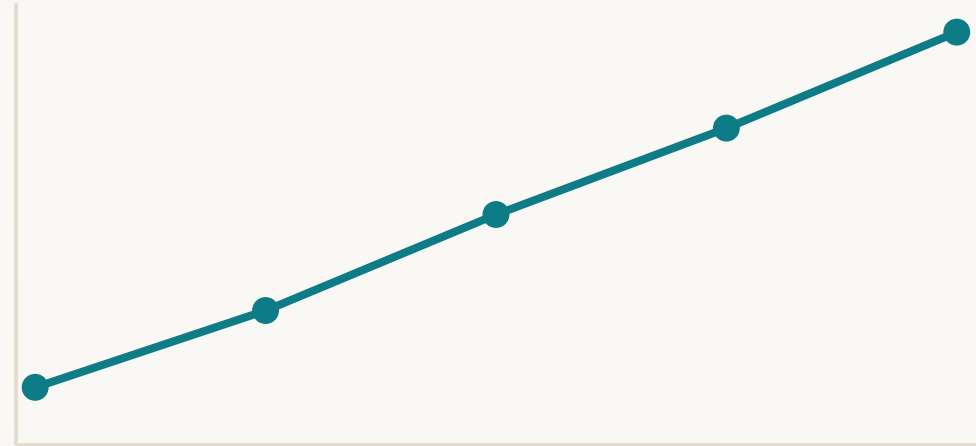
Soft-skills coaching, made a daily habit.

Like Duolingo — but for talking. Track progress, drill weak spots, reward improvement.

SDG 4

Quality Education

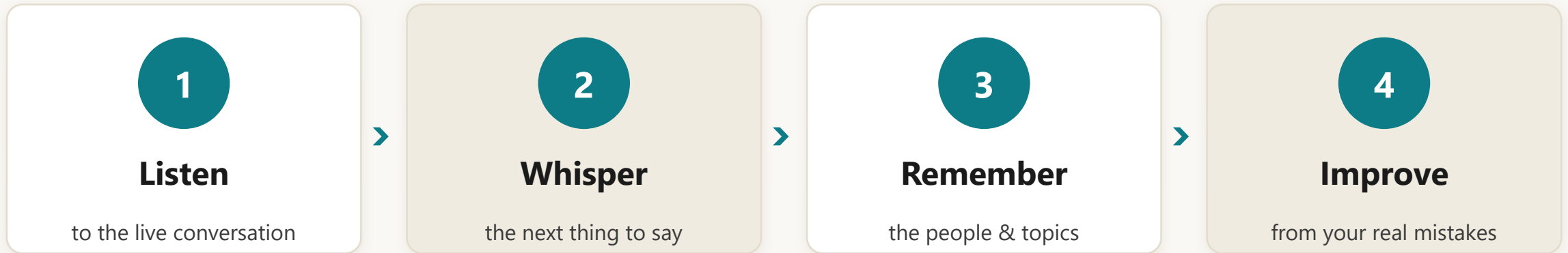
Aligned to the UN Sustainable Development Goal we mapped this project to.



Confidence over time

THE SOLUTION, IN FOUR VERBS

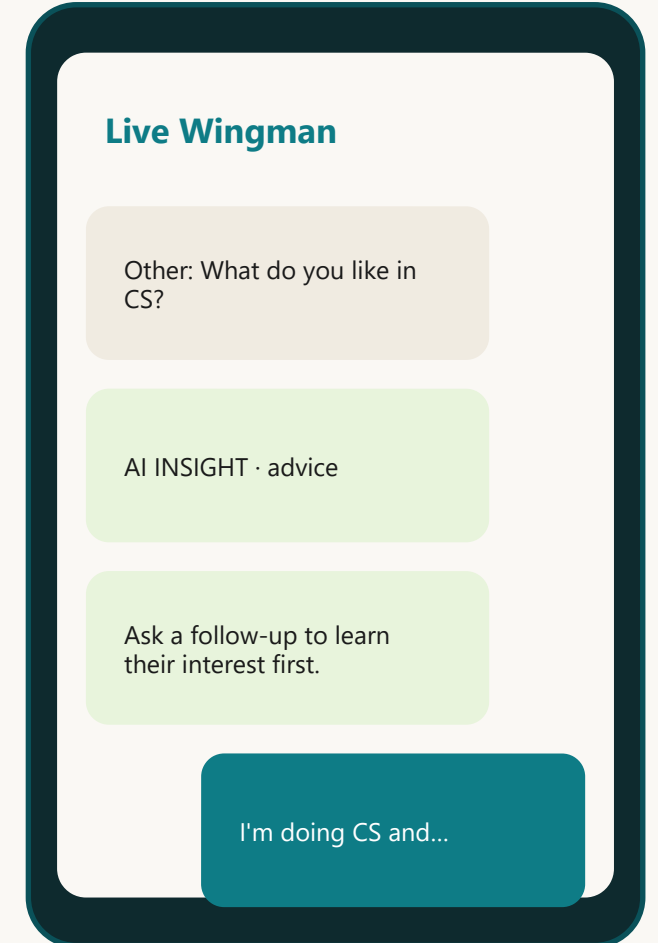
Listen → Whisper → Remember → Improve



MODE 1

Live Wingman — coaches you while you talk.

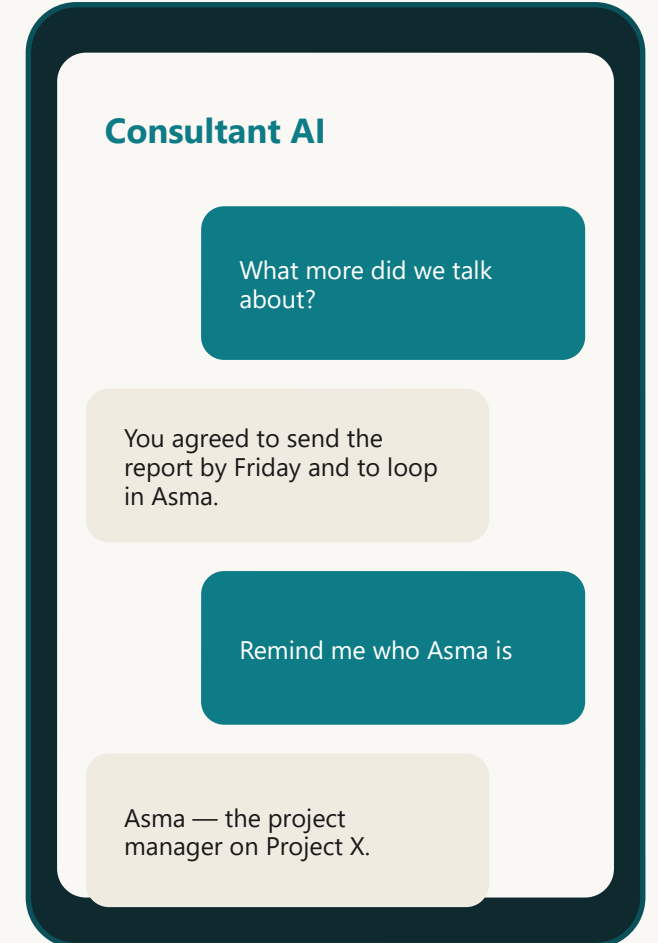
- Real-time advice generated from the live transcript
- Clarifying questions when the moment needs one
- Reacts to the other person — never echoes you back
- Pushed to screen over WebSocket, no flow break



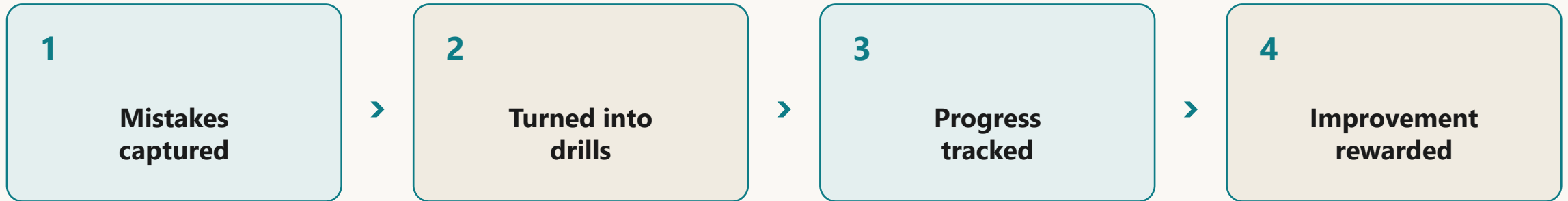
MODE 2

Consultant — ask anything about your past talks.

- Deep, context-aware Q&A over your history
- “What did I promise Client X last week?”
- Streaming, blocking, or batched answers
- Backed by the heavy reasoning model



Speak-Improve — the coaching loop.



↻ *and the loop repeats — every session feeds the next.*

Two memories, not one.

Vector memory

Finds semantically similar past moments — fuzzy recall.

"sounds related"

Knowledge graph

Stores factual relationships between people & topics.

User → knows → Asma

Fused before every answer

Plain RAG sounds right. GraphRAG is right.

✗ Plain vector RAG

Retrieves text that reads similar — can confuse two similar events and invent facts.

✓ Graph-augmented RAG

Looks up hard relationships first (ego-graph around your entities) — anchors the answer.

Thin client. Heavy brain.

The phone

- Microphone
- Screen
- Battery-light



The cloud brain

- All AI: STT, LLM router, memory, graph
- Models stay swappable, not baked into the app
- Same experience on a cheap phone

PART 04

How it actually works.

Three tiers, one clean data path.

Three tiers, top to bottom.

Client — Flutter

Screens · State · Repositories · Cache · Services



Backend — Brain API (FastAPI)

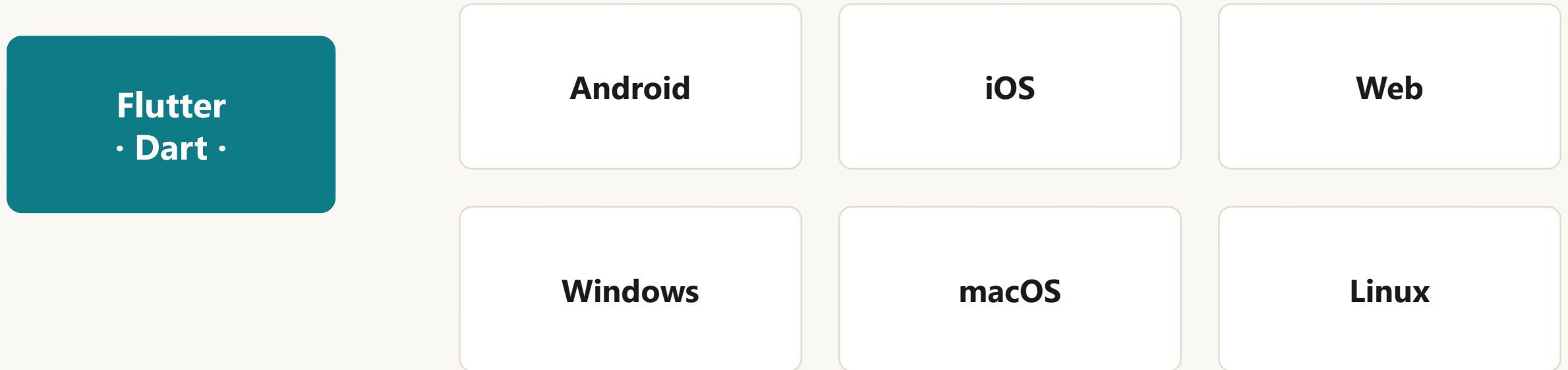
Auth · LLM Router · Endpoints · ARQ Workers



Data & AI

Postgres + pgvector · Redis · Gemini/Cerebras/Groq · LiveKit

One codebase → six platforms.

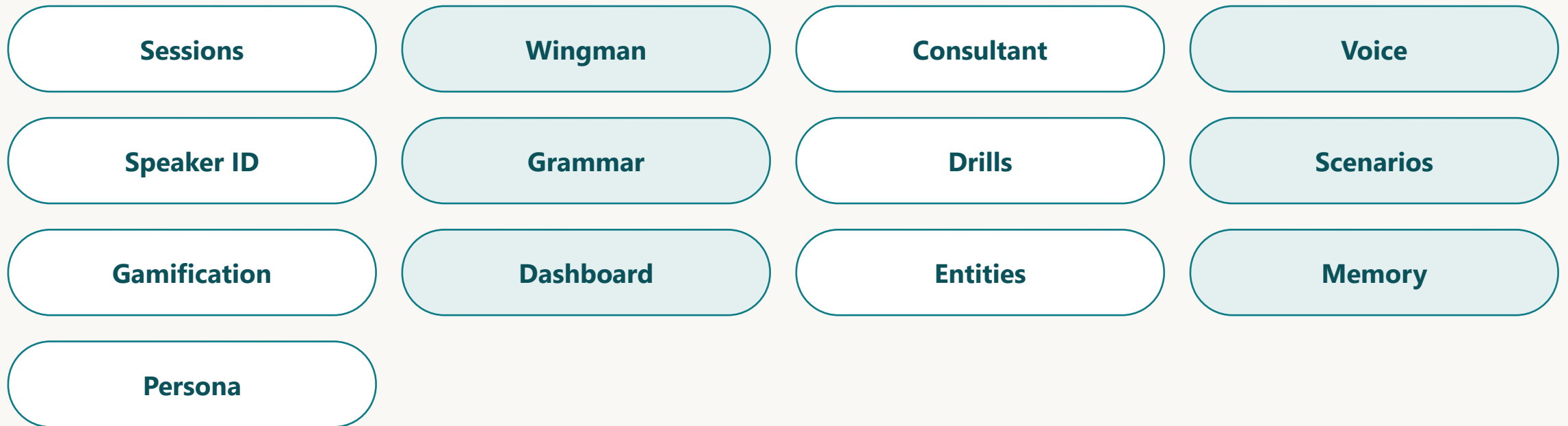


25+ screens · 11 repositories · 25 services · offline-first stale-while-revalidate cache over SQLite

Why not native or React Native?

Approach	Verdict	Why
Flutter	✓ Chosen	One Dart codebase, true desktop, 60fps custom UI
Native (×2)	✗	Two-person team can't build & maintain 6 native apps
React Native	✗	JS-bridge overhead, weak desktop story

Async, top to bottom — 40+ endpoints under /v1.



Python 3.12 · FastAPI on Uvicorn/Gunicorn · Pydantic v2 validates every request & response

Real-time audio needs non-blocking I/O.

✗ Sync (Flask/Django)

One worker blocks per audio stream → doesn't scale.

✓ Async (FastAPI)

Await network/LLM calls without freezing the server. Typed schemas + auto OpenAPI for free.

Each task → the model that fits it.

Task	Model	Provider chain
Consultant Q&A	gemini-2.5-flash	Gemini → Cerebras → Groq
Wingman (live advice)	llama-3.1-8b	Cerebras → Groq → Gemini
Speech-to-text	whisper-large-v3-turbo	Groq
Embeddings	text-embedding-004	Gemini (ONNX fallback)

Per-provider circuit breakers: a failing provider trips its breaker and we fail over instantly.

No single point of failure.

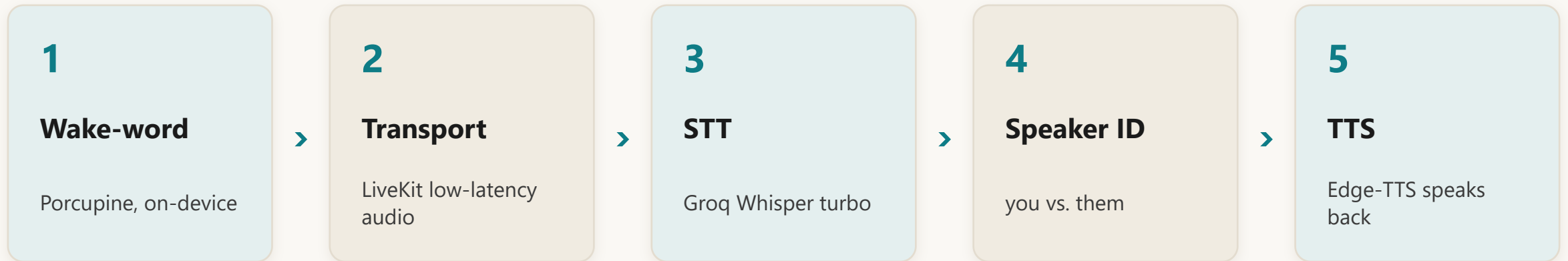
✗ One provider

One outage kills the app. One price hike traps you. One model is wrong for some tasks.

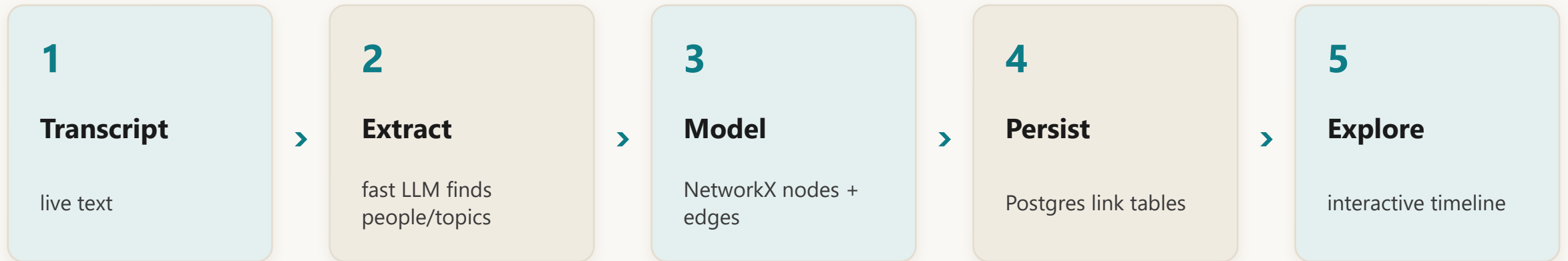
✓ Three providers, routed

Right model per job, automatic failover, no vendor lock-in.

Wake-word → speech → advice → voice.



Transcript → entities → relationships → graph.



Heavy work happens off the critical path.

HOT PATH · stays fast

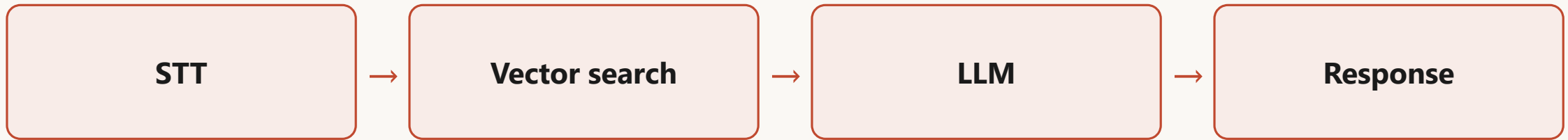
Live advice generation — the only thing on the main thread.

BACKGROUND · 12 idempotent jobs

Embeddings · grammar scan · sentiment · knowledge extraction · rolling summaries · session analytics · drill cards · scenarios · achievements · reminders

THE REAL-TIME CHALLENGE

Chained, it took 3–5 seconds.



Conversation needs

< 1 second. *3–5 s of dead air is useless mid-talk.*

HOW WE BEAT IT

Parallel pipeline + Groq LPU.

Before

~2 s

serial chain, big model



After

~300 ms

hot path only + 8B on Groq LPU

Only advice stays on the main thread; knowledge extraction & memory-saving become fire-and-forget.

CHALLENGE: WHO SAID WHAT

Advise on their words — not yours.

Other Can you explain your hobbies?

You I'm doing CS and... stuff.

Other What do you like in CS?

We parse word-level timestamps to split User vs. Other — Wingman reacts to the other person only.

Hallucination & lost state.

Hallucination

Force graph facts first, plus a strict prompt: "If it's not in the context, say you don't know."

Lost state

Stateless HTTP forgets mid-talk. Hold live context in RAM (fast) and persist to Supabase (reliable).

Production guardrails, not a demo script.



Circuit breakers — isolate a failing provider



Token-bucket rate limiting — Redis Lua, per user



Retries with back-off — Tenacity on transient faults



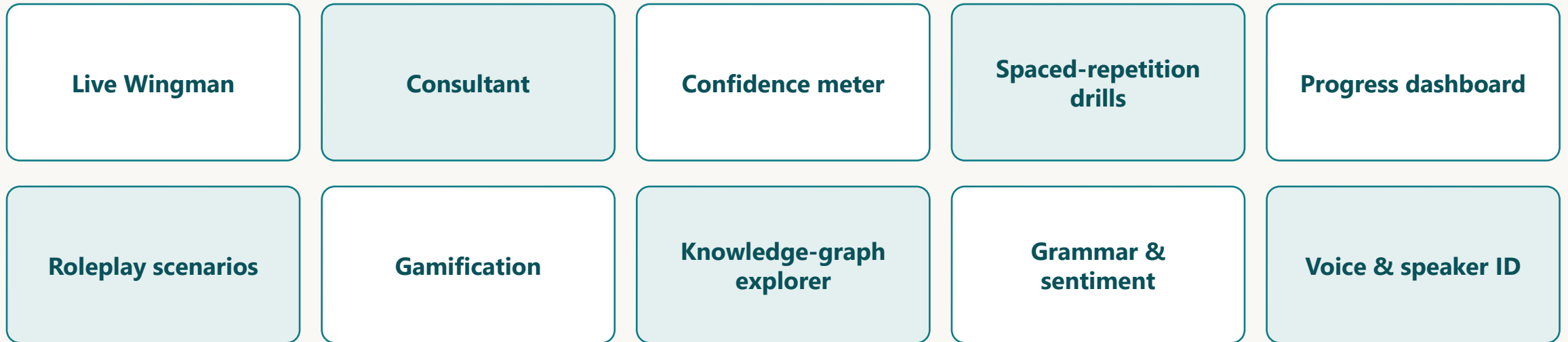
Idempotent jobs — a retry never double-fires



ULID IDs — sortable, collision-safe

WHAT'S BUILT

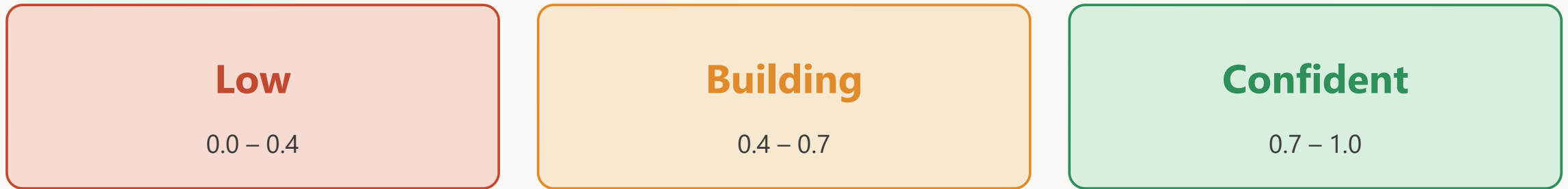
Ten features — all live in the app.



We'll deep-dive four that show real engineering — then a live demo.

On-device, zero added latency.

Counts filler words ("um", "uh", "like") and hedges ("sort of", "I guess") over a rolling 8 seconds — smoothed so it doesn't jitter.



Server only stores the per-turn score at session end — for the confidence-trend chart.

SPACED-REPETITION DRILLS

Your mistakes → flashcards → 5 Leitner boxes.



Correct → up a box (seen less often). Wrong → back to Box 1. XP on every transition (+15 / +5), idempotent.

“Am I actually improving?”

↓ 34%

Mistakes

vs previous window

↑ 35%

Sentiment

vs previous window

+33%

Sessions

vs previous window

+23%

XP

vs previous window

One endpoint, time-bucketed daily / weekly / monthly — pure read-side aggregation, no new tables.

Practice the conversation before you have it.

Salary review with your manager

Person: Asma (Project Manager)

Difficulty: Hard

Opening: *"Thanks for making time — I'd like to talk about my role."*

Generated from your graph

Pass it → auto-graded against its success criteria → +40 XP. The feed never repeats — used topics are excluded.

Habit mechanics — for talking.



Quests — small daily goals



Achievements — milestone badges



Rewards — redeemable points



Streaks — keep the chain alive

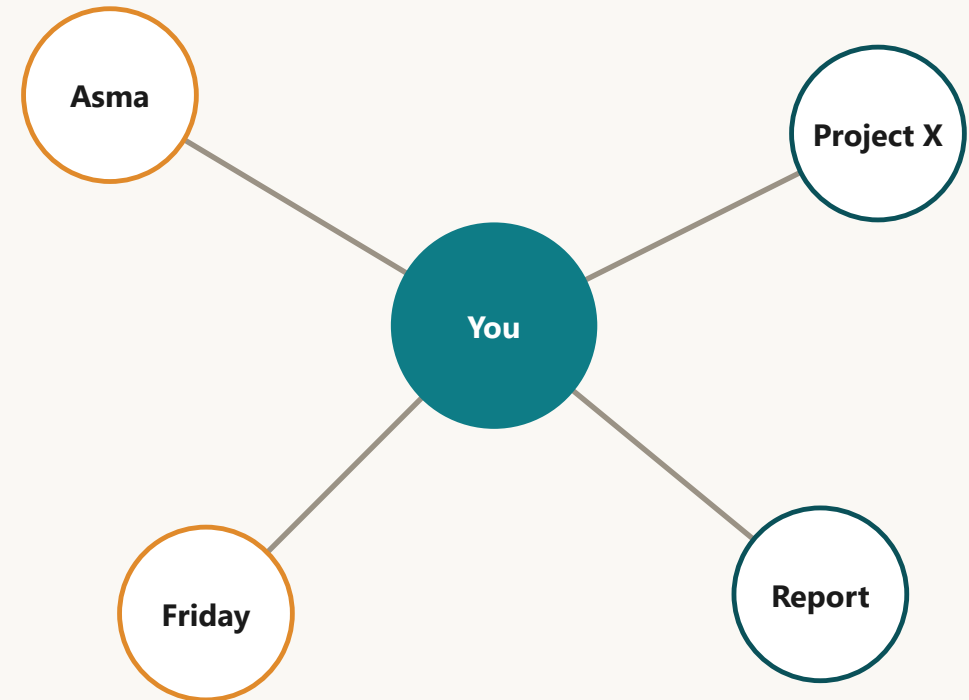


Leaderboard — opt-in only — consistent with privacy-first

Your memory, made tangible.

Browse the people, topics and entities you've mentioned — on a timeline.

The same nodes the AI uses for grounding — now yours to explore.



Three pillars in Postgres.

Vector memory

384-dim embeddings for semantic recall (RAG).

Knowledge graph

Nodes + edges — people, topics, relationships.

Session logs

Verbatim transcript with speaker role.

Supabase Postgres + pgvector · 7 versioned, reversible Alembic migrations.

Your data stays yours.



Row-Level Security

- Enforced at the database
- A user can only ever touch their own graph & memories
- Sharing is opt-in, never default
- Personal growth tool — not org surveillance

We can see the system — from day one.

Sentry

errors

Prometheus

metrics

OpenTelemetry

traces

Grafana

dashboards

structlog

structured logs

HOW WE JUDGE SUCCESS

Honest numbers.

~300 ms

advice latency

$\geq 85\%$

transcription target

low

hallucination rate

~80%

context-accurate answers

Across 30+ trial sessions (~350 transcript lines, 45+ graph relations) — and we're driving accuracy up.

Discipline, not vibes.



mypy --strict — fully typed Python



Ruff — fast linting



pytest — unit · integration · e2e



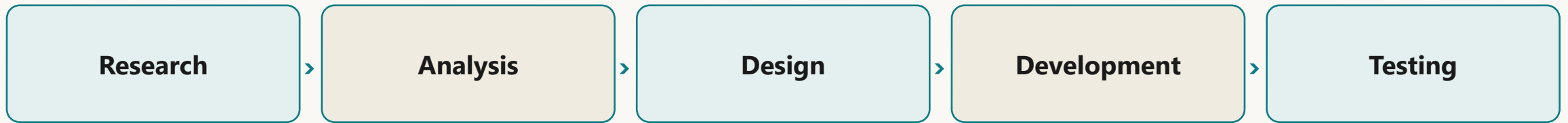
Locust — load tested for concurrency



Docker multi-stage — runtime + worker targets

HOW WE WORKED

Iterative R&D, clean ownership.



Ahmad

Client · architecture · data model · UI · releases

Attique

AI pipeline · memory · backend · workers ·
evaluation

CURRENT SCOPE

The honest edges of what's built today.

English-only

the language we support right now

Needs internet

the AI brain lives in the cloud

Thin client by design

the phone is mic + screen only

Design choices, not loose ends — a clear scope for what we demo today.

The runway.

1

Multi-language

Urdu first, then more

2

Offline fallback

local STT + small LLM (ONNX already in)

3

Tone-aware coaching

live sentiment → tone nudges

4

Open-source release

code + methods for the community

5

Scale & deploy

cloud GPU, app-store builds

6

Fine-tuning

domain-adapt ASR/LLM on sessions



Get better at talking — one conversation at a time.

github.com/qdevaan/Bubbles-AI · qdevaan.github.io/Bubbles-AI

Thank you · Questions?